AQUACULTURE AND SGDS

WHY AQUACULTURE?

BACKGROUND

- Depleted, overharvested or overexploited: If an animal has been hunted beyond sustainable levels;
- Marine/freshwater fisheries are in trouble;
- Increase & appetite: wild fish stocks are shrinking: global pop.;
- FAO 60% of commercial species are either depleted/overfished/fully exploited;
- Deforestation under water;

WHY AQUACULTURE?

- Increase household food supply and improve nutrition.
- Increase household economy through diversification of income and food sources.
- Strengthen marginal economies by increasing employment and reducing food prices.
- To utilise places that cannot be used for other food systems.

(Frankic and Hershner, 2003)

WHY AQUACULTURE?

- Preserve aquatic biodiversity through re-stocking, and recovering of protected species.
- Reduce pressure on fishery resources if done sustainably
- Improving/enhancing habitats
- Earning foreign exchange
- Creating leisure activities e.g. sport, fee fishing, bait

WHY AQUACULTURE?

- Source of income for households;
- Integrated aquaculture also known as agriculture-aquaculture (IAA) is a sustainable way of food production as it uses locally available resources;
- Food security;
- Nutritional security e.g. source of protein;
- Conservation of aquatic species;

WHY AQUACULTURE?

- Source of employment;
- Productive use of poor agricultural land;
- Solution to malnutrition e.g. source of protein;
- Tourism;
- Medicinal properties e.g. fish skin;
- Utilization of water resources e.g. dams, streams and lakes;
- Foreign exchange;
- Contribution to GDP;

AQUACULTURE IN ZAMBIA?

- In Zambia, aquaculture started in the 1940s;
- Growth has been stagnant until early 2000s;
- Aquaculture production increased from 4, 240 tonnes in 2000 to 45, 670 tonnes in 2020 (Hasimuna et al., 2019; FAO, 2022);
- Currently, Zambia is the fifth producer of farmed fish in Africa;
- Cage culture and the intensive land-based production of tilapia;

AQUACULTURE IN ZAMBIA?

Table 2. Top 10 aquaculture producers in Africa in 2018 (FAO 2003-2020a, 2003-2020c, 2004-2020, 2005-2020d, 2005-2020f, 2005-2020c, 2005-2020a, 2005-2020, 2007-2020, 2010-2020b).

No.	Country	Production (metric tons)	Regional share (%)	Global share (%)
1	Egypt	1,561,457	71.10	1.90
2	Nigeria	291,233	13.26	0.35
3	Uganda	103,737	4.72	0.13
4	Ghana	76,630	3.49	0.09
5	Zambia	24,300	1.11	0.03
6	Tunisia	21,756	0.99	0.03
7	Kenya	15,124	0.69	0.02
8	Malawi	9014	0.41	0.01
9	Madagascar	7421	0.34	0.01
10	South Africa	6181	0.28	0.01

Source: Adeleke et al. 2021

AQUACULTURE IN ZAMBIA?

- There is a lot of support from the government through the $50,000,000 loan from the AfDB;
- The Seventh National Development Plan (SNDP) emphasized aquaculture development;
- More farmers have been funded to set up aqua enterprises e.g, hatcheries & growout;
- ZAEDP & CEEC have been spearheading funding through loans to aquaculture entrepreneurs.

AQUACULTURE AND SDGs?

In 2015, the United Nation General assembly presented humanity with a pathway to a more prosperous, equitable, and sustainable future in line with the 2030 Agenda.

Aquaculture is an important sector contributing to human well-being and plays an increasingly important role in efforts to meet the SDGs

(Hambrey, 2017; Troell et al. 2023)

Aquaculture and SDGs?

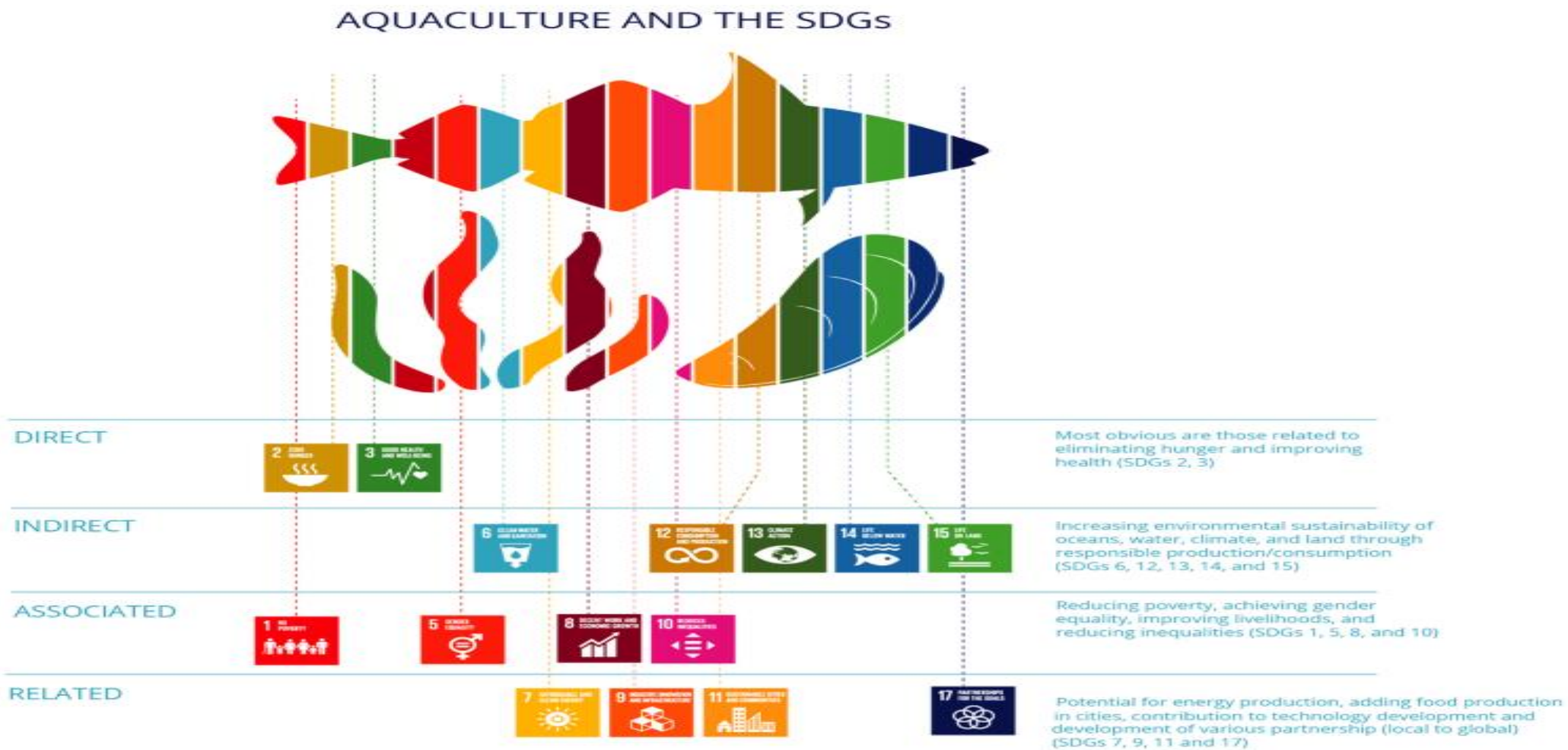


Aquaculture and SDGs?

Aquaculture contributes to all of the SDGs but easy ones to notice are;

- Aquaculture may contribute to all 17 SDGs but the **most obvious** are those related to (A) eliminating hunger and improving health (**SDGs 2, 3**)
- (B) increasing environmental sustainability of oceans, water, climate, and land through responsible production/consumption (**SDGs 6, 12, 13, 14, and 15**)
- (C) reducing poverty, achieving gender equality, improving livelihoods, and reducing inequalities (**SDGs 1, 5, 8, and 10**)

Main contributions to SDGs?



Aquaculture, food security and nutrition provision

What is the definition of food security?

Food security is the measure of an individual's ability to access food that is nutritious and sufficient in quantity (Venugopal,1999).

'4 pillars of food security'.

- **Utilisation** - Not all food is of equal or sufficient value. It is also crucial that individuals have the necessary knowledge and tools to properly 'utilise' the food available to them.

- **Stability** - Good food stability means that access, availability and utilisation of food remains relatively stable over time. It is important to try to minimise any threats to this stability.

Aquaculture, food security and nutrition provision

- **Availability** - Availability simply refers to the existence of food within a community and aquaculture will utilize resources to make seafood accessible.

- **Access** - Simply having enough food in a community means very little if there is poor access to it. Factors such as pricing, household proximity to suppliers and infrastructure all affect our access to food.

Something to think about!

‘’Global hunger is **one of the largest challenges** the world community is faced with. **Up to 828 million people suffer from hunger** (status as of 2022). One child dies of hunger every ten seconds”

Aquaculture, food security and nutrition provision

What is the definition of nutrition security?

Nutrition security demands the intake of a wide range of foods which provides the essential needed nutrients (Venugopal,1999).

AQUACULTURE CONTRIBUTION TO FOOD AND NUTRITION SECURITY

- Aquaculture plays a central role in food security and nutrition for billions of people and constitutes a cornerstone of many people's livelihoods, economies, and cultural practices.

- Global aquaculture production has tripled in live-weight volume from 34 M in 1997 to 112 Mt in 2017

Aquaculture, food security and nutrition provision

- aquaculture related activities also support the livelihoods of more than 120 million people worldwide, the majority of whom live in economically developing countries.
- Aquaculture helps to reduce the price of fish which increase access and opportunities for income
- aquaculture is an important mechanism for local food security by reduce vulnerability to uncontrollable natural crashes in aquatic production, improved food availability, improved access to food and more effective food utilization

Simple question

1. Nutrition security demands the intake of a wide range of foods which provides the essential needed nutrients. **True or False**

2. Aquaculture is defined as the farming of livestock species like goats and pigs. **True of False**

3. With the reduction of seafood from natural water bodies, aquaculture is seen as a major source of cheap protein source and a good contributor to seafood availability. A) Yes B) Maybe C) No

4. Which of the following are the SDGs that aquaculture supports directly. A) Zero hunger B)Life below water C) Innovation and Infrastructure

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